

DAYTON INTL, OH, USA

WMO: 724290

Lat: 39.906N

Lon: 84.219W

Elev: 306

StdP: 97.71

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93815

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.8	-13.6	-20.9	0.6	-16.0	-18.2	0.8	-13.0	13.0	-1.0	12.0	-0.8	4.2	270	0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	32.3	23.2	31.0	22.7	29.7	21.9	24.8	30.0	24.0	28.8	23.2	27.7	4.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.1	18.6	27.7	22.4	17.8	26.8	21.7	17.0	26.0	76.5	29.9	73.3	28.9	70.4	27.7	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.2	9.9	8.7	-20.1	34.1	3.9	1.9	-22.9	35.4	-25.2	36.5	-27.4	37.6	-30.2	38.9
			-20.4	26.3	3.8	1.1	-23.2	27.1	-25.4	27.7	-27.6	28.3	-30.3	29.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.6	-2.3	-0.3	4.8	11.5	17.1	21.9	23.6	22.9	19.3	12.8	6.1	0.7
	DBStd	10.45	7.03	6.49	6.29	5.26	4.69	3.45	2.92	2.84	4.04	4.98	5.39	5.93
	HDD10.0	1376	386	292	186	44	4	0	0	0	0	28	142	294
	HDD18.3	3023	641	521	419	212	82	9	1	2	37	183	368	548
	CDD10.0	1947	4	4	26	88	224	358	421	400	278	114	24	5
	CDD18.3	552	0	0	1	7	44	118	163	144	64	11	0	0
	CDH23.3	4492	0	0	8	51	339	932	1388	1154	540	80	1	0
	CDH26.7	1377	0	0	1	4	73	294	471	355	164	16	0	0
Wind	WSAvg	4.3	5.2	5.0	4.9	5.0	4.3	3.8	3.4	3.2	3.4	4.1	4.6	4.8
Precipitation	PrecAvg	980	65	55	86	99	105	101	94	78	75	67	78	75
	PrecMax	1518	234	147	194	221	228	242	217	204	275	159	205	255
	PrecMin	615	8	6	27	14	39	8	12	0	7	4	18	15
	PrecStd	195	37	31	38	45	50	44	40	46	54	37	39	37
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.6	19.4	24.3	27.2	30.4	33.1	34.7	33.2	32.9	28.8	21.5	17.5
		MCWB	14.1	14.6	16.0	17.7	21.1	23.3	24.9	23.5	21.1	20.1	15.2	14.8
	2%	DB	14.0	15.1	20.8	24.8	28.6	31.3	32.3	31.6	30.7	25.5	18.8	14.6
		MCWB	12.5	11.5	14.1	16.3	20.0	22.5	23.9	23.1	21.2	18.2	14.0	12.3
	5%	DB	11.1	12.3	17.7	22.6	27.1	29.8	30.8	30.2	28.5	23.1	16.9	11.9
		MCWB	9.4	9.4	12.6	15.0	19.5	21.6	23.1	22.5	20.4	17.2	13.1	9.9
	10%	DB	7.8	9.3	14.9	20.3	25.2	28.4	29.3	28.7	26.5	21.0	14.5	9.3
		MCWB	5.7	6.3	10.8	13.9	18.6	21.0	22.3	21.7	19.3	16.3	11.1	7.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.7	15.3	16.9	19.5	22.8	25.1	26.2	25.3	23.9	21.8	17.1	15.2
		MCDB	16.0	18.5	22.5	25.1	27.8	30.4	32.1	30.7	29.2	26.8	19.6	17.0
	2%	WB	12.5	12.4	15.0	17.7	21.6	23.9	25.1	24.3	22.7	20.1	15.3	12.8
		MCDB	13.9	14.3	19.3	22.4	26.8	29.1	30.6	29.2	27.7	23.3	17.8	14.1
	5%	WB	9.8	9.5	13.4	16.4	20.5	23.0	24.2	23.7	21.7	18.4	13.5	10.2
		MCDB	11.1	12.3	16.8	20.5	25.2	28.1	29.0	28.1	26.3	22.1	16.2	11.6
	10%	WB	5.8	6.4	11.2	15.0	19.5	22.1	23.4	22.9	20.7	16.7	11.6	7.4
		MCDB	7.4	9.1	14.6	19.3	23.8	26.7	27.8	26.9	24.5	20.4	14.2	9.2
Mean Daily Temperature Range	5% DB	MDBR	7.9	8.5	9.6	11.2	10.3	9.9	9.9	10.2	11.1	10.3	8.9	7.2
		MCDBR	9.7	11.9	12.5	13.2	11.8	11.4	10.8	11.2	12.9	12.6	11.2	10.0
		MCWBR	8.5	9.4	7.8	7.6	5.7	5.1	4.4	4.5	5.4	6.8	8.0	8.7
	5% WB	MCDBR	9.2	11.3	11.1	11.3	10.1	9.8	9.8	9.4	10.3	10.9	10.0	9.4
		MCWBR	8.6	9.8	7.8	7.5	5.4	4.9	4.5	4.4	5.1	6.6	8.2	8.9
Clear-Sky Solar Irradiance	taub	0.300	0.323	0.343	0.391	0.435	0.442	0.445	0.439	0.392	0.353	0.322	0.299	
	taud	2.441	2.351	2.373	2.253	2.174	2.202	2.204	2.203	2.312	2.418	2.462	2.464	
	Ebn at Noon	869	892	911	883	848	839	833	827	846	848	834	841	
	Edh at Noon	80	101	110	133	147	143	142	138	116	94	78	73	
All-Sky Solar Radiation	RadAvg	1.70	2.57	3.56	4.79	5.49	6.07	5.96	5.46	4.46	3.12	2.09	1.43	
	RadStd	0.15	0.30	0.31	0.40	0.46	0.41	0.35	0.27	0.41	0.29	0.25	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+107	+60
(47)	Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+112	+65

Nomenclature: See separate page